

Complements of Operational Research

3.3 Sequencing problems: the time deviation method (TDM)

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A heuristic for sequencing

A **heuristic** is a fast rule that gives a reasonable solution, **with no guarantee** of being optimal.

The **TDM** (*time deviation method*, Rao, Raju & Babu, 2013) decides the order by looking, for each time, at two deviations:

- **machine deviation** = (largest time on that machine) – (time);
- **job deviation** = (largest time of that job) – (time).

Each cell of the table gets a pair (machine deviation, job deviation).

Cell (0, 0)

The time is the **largest on its machine** and the **largest of its job**.

- on machine 1: a “heavy” job in printing → should go to the **end** of the sequence;
- on machine 2: a “heavy” job in finishing → should go to the **start**.

It is the same intuition as Johnson’s, but applied to the *largest* times instead of the smallest.

The algorithm (two machines)

TDM

1. Compute the deviation table for the jobs not yet assigned.
2. Jobs with a (0, 0) cell on **machine 1** enter the sequence under construction **from the front** (after those already there).
3. Jobs with a (0, 0) cell on **machine 2** enter **from the back** (before those already at the end).
4. If there are several (0, 0) cells on the same machine, first the one with the largest sum of the job's deviations.
5. Go back to 1 until all are assigned. At the end, **reverse** the sequence built.

With two machines there is always at least one (0, 0) cell: the largest of all the times not yet assigned is the largest on its machine and of its job.

m machines: as in 3.2, reduce to $G = M_1 + \cdots + M_{m-1}$, $H = M_2 + \cdots + M_m$ and apply the TDM to (G, H) .

TDM in the print shop: iteration 1

Job	times		deviations (machine, job)	
	Print.	Finish.	Print.	Finish.
A	8	9	(3, 1)	(3, 0)
B	11	5	(0, 0)	(7, 6)
C	5	2	(6, 0)	(10, 3)
D	3	5	(8, 2)	(7, 0)
E	4	12	(7, 8)	(0, 0)
F	10	9	(1, 0)	(3, 1)
max.	11	12		

Examples of the computation:

- A in printing: $11 - 8 = 3$ and $9 - 8 = 1 \rightarrow (3, 1)$;
- C in finishing: $12 - 2 = 10$ and $5 - 2 = 3 \rightarrow (10, 3)$.

(0, 0) cells:

- **B** in printing $\rightarrow$ enters from the front;
- **E** in finishing $\rightarrow$ enters from the back.

Sequence under construction:

B	_	_	_	_	E
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Iterations 2 and 3 and the reversal

Iteration 2 (A, C, D, F; maxima 10 and 9)

Job	Print.	Finish.
A	(2, 1)	(0, 0)
C	(5, 0)	(7, 3)
D	(7, 2)	(4, 0)
F	(0, 0)	(0, 1)

F from the front, A from the back: B F _ _ A E

Iteration 3 (C, D; maxima 5 and 5)

Job	Print.	Finish.
C	(0, 0)	(3, 3)
D	(2, 2)	(0, 0)

C from the front, D from the back: B F C D A E

Reverse

Sequence built: B, F, C, D, A, E

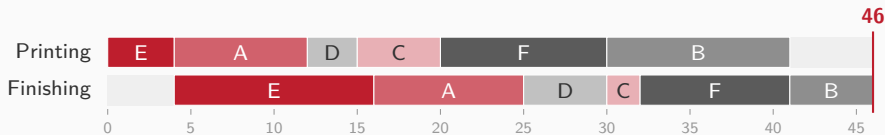
TDM sequence: E, A, D, C, F, B

	E	A	D	C	F	B
C_1	4	12	15	20	30	41
C_2	16	25	30	32	41	46

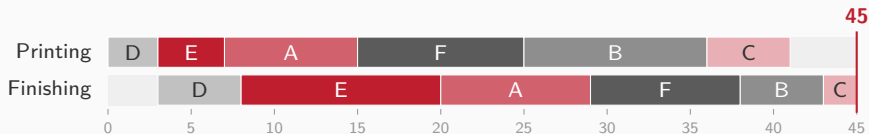
$T = 46$ h: much better than the order of arrival (53 h), but 1 h above the optimum (45 h, Johnson). Of the 720 sequences, 40 give 46 h or less.

Johnson and TDM side by side

TDM: E, A, D, C, F, B $\Rightarrow T = 46$



Johnson: D, E, A, F, B, C $\Rightarrow T = 45$



The TDM starts with the job with the longest finishing (E); Johnson starts with the one with the shortest printing (D), and finishing gets going 1 h earlier. The TDM ends with B (5 h of finishing); Johnson with C (2 h).

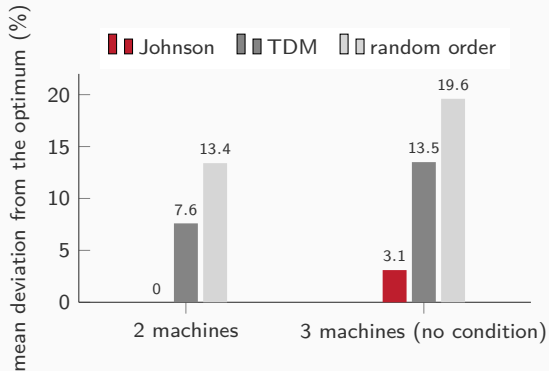
The three examples of the chapter

	Order of arrival	TDM	Johnson	Optimum
Print shop, 2 machines (3.1)	53	46	45	45
With cutting, 3 machines, condition holds (3.2)	55	49	47	47
Laminating, 3 machines, condition fails (3.2)	39	40	39	35

total times in hours; optimum by brute force; in the last case, TDM and Johnson on (G, H)

- When the condition holds, Johnson already gives the optimum — the TDM adds nothing.
- When the condition fails, both rules fail: 40 h and 39 h against 35 h (the order of arrival also gives 39 h). An **exact** method that does not enumerate everything is needed → 3.4.

Numerical experiment: 500 random problems



7 jobs, integer times from 1 to 20, optimum by brute force (5040 sequences).

% of times optimal	2 mach.	3 mach.
Johnson	100	49
TDM	18	3
random	8	1

Worst case of the TDM: 27% (2 mach.) and 35% (3 mach.) above the optimum.

Reading: the TDM is better than sequencing at random but, in these tests, worse than Johnson — even when Johnson is no longer exact.

Example 6 — Example 1 of 3.1 with the TDM

Apply the TDM to the nine jobs of Example 1 (3.1). Compare the sequence and the total time with those of Johnson.

Example 7 — Example 5 of 3.2 with the TDM

Apply the TDM to the five machines of Example 5 (3.2), after reducing them to (G, H) , and compare with Johnson.

Example 8 — three machines

Job	1	2	3	4	5
M1	7	8	10	9	7
M2	5	6	4	5	7
M3	12	10	7	8	11

Check the condition, apply the TDM and Johnson, and compare the total times.

KEY TAKEAWAYS — 3.3

1. Deviations of each time: (largest on the machine – time, largest of the job – time).
2. $(0, 0)$ cell on machine 1 $\rightarrow$ enters from the front; on machine 2 $\rightarrow$ enters from the back; recompute; at the end **reverse**.
3. With m machines, apply the TDM to the fictitious machines (G, H) .
4. The TDM is a **heuristic**: always compute the total time and compare (with Johnson, or with the optimum when it is known).

Next: 3.4 *Branch and bound* — the optimum without enumerating everything

References Full list: *Course bibliography*.

- Rao, N. N., Raju, O. N., Babu, I. R., Modified heuristic time deviation technique for job sequencing and computation of minimum total elapsed time, *International Journal of Computer Science & Information Technology*, 5(3), 67–77, 2013. doi:10.5121/ijcsit.2013.5305.
- Kasana, H. S., Kumar, K. D., *Introductory Operations Research*, Springer, 2004 — ch. 10.
- Pinedo, M. L., *Scheduling: Theory, Algorithms, and Systems*, 5th ed., Springer, 2016 — heuristics for *flow shops*.

Resources

- Exercise sheet 3. `code/python/03_scheduling/03_3_tdm/desvio_tempo.py` (function `tdm`).